Arkaprava Saha

Curriculum Vitae

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Webpage

Education

2018–Present **Doctor of Philosophy (PhD) in Computer Science and Engineering (CSE)**, *Nanyang Technological University (NTU)*, Singapore, Data Management and Analytics Laboratory (DMAL).

- **Thesis:** Efficient Optimization of Network Metrics in Large Uncertain Graphs
- **Advisor:** Prof. Arijit Khan (till April 2022), Prof. Cheng Long (after April 2022)
- **Graduate Courses:** Information Retrieval and Analysis, Data Mining, Deep Learning for Data Science, Advanced Topics in Distributed Systems, Artificial Intelligence in Game Design, Database Systems
- **CGPA:** 4.0/5.0

2014–2018 **Bachelor of Technology (BTech) in Computer Science and Engineering (CSE)**, *Indian Institute of Technology (IIT) Delhi*, New Delhi, India, Departmental Specialization in Data Analytics and Artificial Intelligence.

- **Thesis:** Random Walks and Reachability in Large Bloom Graphs
- **Advisor:** Prof. Maya Ramanath
- **Graduate Courses:** Machine Learning, Logic For Computer Science, Natural Language Processing, Numerical Algorithms
- **CGPA:** 8.187/10.0, **DGPA:** 8.223/10.0

Experience

May 2017 – July 2017 **Software Engineering Intern**, *Samsung R&D Institute*, Bangalore, India, Android Platform Team.

- **Mentor:** Preeti Rani Deka
- **Description:** Designed a generalized parser for any arbitrary watch face (dial) of the Samsung GWD to be rendered on an Android phone, thereby improving the old parser which was completely hardcoded for some specific watch faces only.

Publications

- **Most Probable Densest Subgraphs**. Arkaprava Saha, Xiangyu Ke, Arijit Khan, Cheng Long. (Under Review) *International Conference on Data Engineering (**ICDE**) 2023.*
- **Voting-based Opinion Maximization**. Arkaprava Saha, Xiangyu Ke, Arijit Khan, Laks V.S. Lakshmanan. *International Conference on Data Engineering (**ICDE**) 2023.*
- **Shortest Paths and Centrality in Uncertain Networks**. Arkaprava Saha, Ruben Brokkelkamp, Yllka Velaj, Arijit Khan, Francesco Bonchi. *International Conference on Very Large Data Bases (**VLDB**) 2021.*
- **Reachability in Large Graphs using Bloom Filters**. Arkaprava Saha, Neha Sengupta,

Maya Ramanath. *International Conference on Data Engineering (**ICDE**) 2019 Workshop on Large Scale Graph Data Analytics.*

Research Interests

The broad area of big graph data mining and management; currently focusing on algorithms for efficient approximation of network metrics in large complex graphs (e.g. uncertain or probabilistic graphs, social networks); open to research in other areas of big graph data as well

Recent Research Projects

Densest Subgraphs in Uncertain Graphs

Collaborators: Xiangyu Ke (ZJU), Arijit Khan (AAU), Cheng Long (NTU)

Description: This project involves finding, in an uncertain or probabilistic graph, the node set that is the most likely to induce a densest subgraph. **This project has culminated in a research paper which is currently under review.**

Services

Program Committee (PC) Member

2022 EAI ICECI

2021 SEA Data @ VLDB

External Reviewer

2023 ICDE

2022 VLDB, SIGMOD, SIGKDD

2021 ICDE, VLDB

2020 SIGMOD

2019 VLDB, WWW, CIKM, DSAA

Volunteering

2022 Interview with Prof. Hanwang Zhang, winner of the Singapore Young Scientist Award 2021

2020 VLDB Student Volunteer

Positions of Responsibility

2019-2020 President of NTU School of CSE Graduate Students' Club

Talks

- **Shortest Paths and Centrality in Uncertain Networks**. *International Conference on Very Large Data Bases (**VLDB**) 2021.*

Technical Skills

Languages: C, C++, Python, Java, SQL, SML

Software: MATLAB, LaTeX, Android Studio

Teaching

Teaching Assistant (TA)

NTU Introduction to Databases (2018-19 Semester 1, 2019-20 Semesters 1 & 2, 2020-21 Semester 1)

NTU Data Structures (2018-19 Semester 2)

NTU Data Structures and Algorithms (2020-21 Semester 2)

Honors and Awards

2021 **VLDB Conference Attendance Support Award**, *Very Large Data Base (VLDB) Endowment*, United States of America.
Awarded the grant for attending the conference

2019 **University Teaching for Teaching Assistants**, *Nanyang Technological University (NTU)*, Singapore.
Officially certified by the university as "Recommended to Teach"

2018 **Commonwealth Research Scholarship**, *Nanyang Technological University (NTU)*, Singapore.
Awarded the scholarship for pursuing PhD

2014 **JEE Advanced (IIT-JEE)**, *Indian Institutes of Technology (IITs)*, India.
Secured **All India Rank 98** out of around 1.3 million students and **State Rank 1**

2014 **West Bengal Joint Entrance Examination (WBJEEM)**, *West Bengal Joint Entrance Examinations Board (WBJEEB)*, India.
Secured **State Rank 6**

2013 **Kishore Vaigyanik Protsahan Yojana (KVPY)**, *Department of Science and Technology (DST)*, Government of India.
Selected as a **Fellow** in the SA stream with **All India Rank 56** out of around 0.15 million candidates

2012 **Regional Mathematical Olympiad (RMO)**, *Homi Bhabha Centre for Science Education (HBCSE)*, India.
Among 35 students in West Bengal to qualify for the national level examination

2010 **National Talent Search Examination (NTSE)**, *National Council of Educational Research and Training (NCERT)*, India.
Selected as a **Scholar**

Languages

Bengali	*Mother Tongue*
English	*Academic and Professional*
Hindi	*Basic Conversations*